

Deliverable D4.1

Updated Scientific Plan for Local and Global Autonomous Motion Planning

DRAFT – VERSION 0.1
Planning for Artificial Robotic Intelligence and Beyond

Project acronym	Plan4ARI
Project duration	48 months
Work package/activity	Activity 4 – Local and Global Autonomous Motion Planning
Task basis	Task 4.1 Update of the Scientific Plan with the latest State-of-the-Art Methodologies and Tools for Motion Planning; Task 4.7 Benchmarking and Evaluation
Activity period	M1–M6
Suggested delivery point	M6 (to be confirmed)
Document status	Version 0.1
Lead beneficiary	CNR
Dissemination level	To be confirmed

Document control

Version	Date	Author/reviewer	Change description
0.1	20 July 2026	TBC	Initial draft outline
0.2	1 October 2026	N. Pedrocchi (CNR)	State of the art (MPPI, path-wise redundancy, base placement), revised architecture, benchmark plan
1.0	TBC	TBC	Approved deliverable

Contents

1	Executive summary	5
2	Objectives, scenarios and baseline architecture	6
2.1	Objectives of Activity 4	6
2.2	Baseline architecture	6
2.3	Reference application	6
2.4	Requirements	6
3	Sampling-based optimal planning: state of the art	8
3.1	Speed of sampling-based planners	8
3.2	Tolerance-aware Cartesian path planning	8
3.3	Task localisation and base placement	8
3.4	Sampling-based planners guiding local optimisers	8
4	Dynamic replanning and human-aware safety	9
4.1	Safety-aware objectives	9
4.2	Reactive, risk-aware and constraint-aware sampling	9
4.3	Implications	9
5	Model predictive control and MPPI for trajectory generation	10
5.1	Theoretical framework	10
5.2	MPPI versus MPC and other horizon-based planners	10
5.3	Analysis of the literature	11
5.4	Industrial look-ahead, path following and micro-interpolation	13
6	Software frameworks and industrial toolchain	14
7	Gap analysis and revised architecture	15
7.1	What changes with respect to the proposal	15
7.2	Updated approach: sampling-based planning and MPPI	15
7.3	End-to-end pipeline	15
7.4	Open IPC / Core IPC allocation	16
7.5	Robot-agnostic design	17
8	Selected methodologies and fallback options	18
9	Benchmarking and evaluation plan (Task 4.7)	19
9.1	Application catalogue	19
9.2	Protocol and KPIs	19
9.3	Benchmark repository	20

10 Integration roadmap, risks and compliance **21**

10.1 Mapping onto Tasks 4.2–4.7 21

10.2 Risks 21

10.3 Standards and compliance 21

11 Conclusions and approved updates **22**

11.1 Retained and modified methods 22

11.2 Approved benchmark and integration plan 22

11.3 Open scientific questions and technology watch 22

11.4 Consortium approval record 22

1 Executive summary

D4.1 updates the scientific plan of Activity 4 with the state of the art that emerged after the submission of [Plan4ARI](#). The original pillars — informed sampling-based planning with multi-path reuse (T4.2) and MPC micro-interpolation in the Core IPC (T4.3) — are **retained**; an **MPPI-based look-ahead layer is added** between them, and the end-to-end pipeline is made explicit, starting from the **optimal localisation of the robot with respect to the task**.

Since the proposal, sampling-based motion planning has become fast enough to answer collision-free queries in microseconds on a standard CPU [1], and Model Predictive Path Integral (MPPI) control has matured into a practical, derivative-free receding-horizon method for manipulators and mobile robots, with open implementations and industrial adoption in the ROS 2 navigation stack [2–4]. A review of 119 sources (Sections ??–6) led to the following decisions:

- **Sampling-based planning and MPPI.** MPPI is adopted as a *stochastic look-ahead* that explores the task redundancy (tool roll, orientation tolerance, external axes, base) ahead of the robot, handles discrete configuration alternatives as multiple goals and reacts to scene changes. Sampling-based planning remains in charge of transfer motions, station-level feasibility and multi-path reuse (Section 7).
- **Explicit pipeline.** Scene model → task segmentation → optimal task localisation (stations, base, rail) → segment feasibility and cost-to-go → online execution (MPPI + transfers) → Core-IPC path-velocity MPC.
- **Open IPC / Core IPC.** All stochastic computation runs in the Open IPC; the Core IPC receives smooth references and enforces hard limits and safety rules deterministically at 1 kHz.
- **Robot-agnostic framework.** Kinematic and task descriptions are inputs; inverse-kinematics solution classes generalise the classical eight solutions of spherical-wrist arms to cuspidal arms, extended joint ranges, external axes and mobile bases.
- **Benchmarking (T4.7).** A catalogue of 17 applications; arc welding of large structures (reference demonstrator) is validated on real hardware, **all other applications are benchmarked in simulation**.

The expected effects are a reduction of reconfigurations and safety-induced idle time, constant process speed within tolerance, and portability across robot kinematics, while keeping the computation-time KPI comparable with the OMPL baseline of the proposal (CPU-first design).

2 Objectives, scenarios and baseline architecture

2.1 Objectives of Activity 4

Activity 4 develops motion planners for teams of mobile manipulators and industrial arms working in dynamic, human-shared environments, integrated in the COMAU MI.RA/Dexter platform and in the Core IPC. The project targets are autonomous plan generation in all working conditions (joint and Cartesian), a reduction of about 30% of the execution time of interaction tasks with respect to state-of-the-art planners, and safe adaptation of the velocity to the working conditions (Project Description, §A.4.4).

2.2 Baseline architecture

The new generation of controllers relies on two computing units (Figure 1). The *Open IPC* hosts the programming tools (MI.RA/Dexter, ROS 2), the sensing and the motion planner, and acts as the bridge between the non-real-time and the real-time worlds. The *Core IPC* runs the hard real-time motion control, the micro-interpolation (high-rate real-time motion planner) and all safety rules. The original plan combined informed sampling-based planning in the Open IPC with an MPC micro-interpolator in the Core IPC.

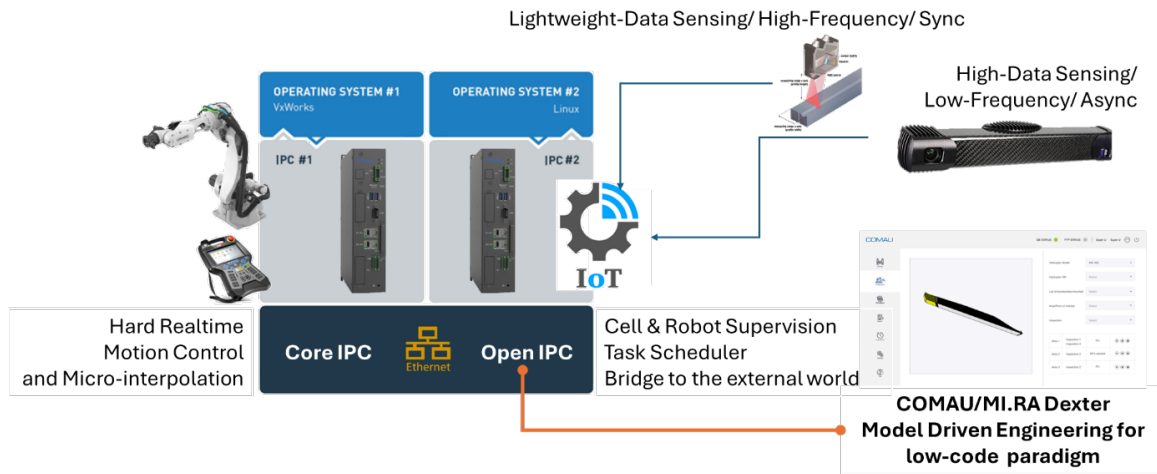


Figure 1 Baseline architecture of the new generation of robot controllers (from the Project Description): the Open IPC provides programming tools, motion planner and the bridge to cell and sensing; the Core IPC oversees control loops, micro-interpolation and safety rules.

2.3 Reference application

Arc welding of very large metal structures by an industrial arm mounted on an AGV (optionally on an interpolated linear axis). The AGV is placed at a station and the arm welds a portion of the structure; base motion interpolated with the arm during welding is a secondary, theoretically relevant option. The task is *under-constrained*: the seam point must be reached exactly, the torch axis may lie within a cone of about 10° around the nominal direction (possibly anisotropic between work and travel angle), the rotation about the torch axis is almost always free, and the travel speed must stay constant within $\pm 10\%$ along the whole seam. Because of the size of the structure the arm may have to change configuration: stop, retract, reconfigure, approach and restart about 1 cm before the detach point. Seam segmentation and a raw positioning of the manipulator that reduces reconfiguration risk are standard industrial practice and are made explicit in the updated pipeline.

2.4 Requirements

- R1. **Exact task, free redundancy.** Task constraints (position, axis) are hard; tolerances (cone, roll, external axes, base) are exploited as optimisation freedom [5, 6].
- R2. **Speed band.** Travel speed within a fixed band ($\pm 10\%$) under joint velocity, acceleration and jerk limits [7].
- R3. **Planned reconfigurations.** Fewest stop-reconfigure-restart cycles, with planned transfer motions and restart overlap [8, 9].

- R4. **Optimal task localisation.** Station / base / rail placement chosen so that whole task segments are feasible [10, 11].
- R5. **Registration.** Mobile platforms on large workpieces show centimetre-level errors without local registration [12]; plans are executed on the locally registered task (seam scan [13], Activity 3).
- R6. **Human-shared space.** ISO/TS 15066-aware costs and certified speed adaptation [14, 15].
- R7. **Robot-agnostic.** Any serial arm (non-cuspidal or cuspidal [16]), joints with range beyond $\pm\pi$, external linear/rotary axes, mobile bases.
- R8. **Industrial integration.** Deterministic, repeatable outputs; dockerised actions in MI.RA/Dexter; hard real-time Core IPC.

3 Sampling-based optimal planning: state of the art

To be completed by the T4.2 team: informed sets, admissible informed and local sampling, adaptive exploration/exploitation, anytime and multi-query planning with path reuse (Project Description, §A.4.2). The subsections below summarise the elements of the updated review that affect the revised architecture.

3.1 Speed of sampling-based planners

Vectorising forward kinematics and collision checking with CPU SIMD instructions accelerates any sampling-based planner by more than two orders of magnitude, with median planning times of tens of microseconds for a 7-DoF arm on a single core and support for low-power ARM boards [1]. GPU-parallel motion generation (parallel IK, geometric planning, particle-based seeding and L-BFGS) plans collision-free arm motions in tens of milliseconds [17]. For Plan4ARI this makes sampling-based planning usable inside the online loop, e.g. to plan transfer motions for reconfiguration at every cycle, and to evaluate station candidates exhaustively.

3.2 Tolerance-aware Cartesian path planning

Industrial processes (welding, spraying, machining) prescribe a Cartesian path with tolerances. The reference open-source approach, Descartes, samples each path point within its tolerances, computes all IK solutions and searches a layered graph; its evaluation on arc welding documents memory growth, the absence of any speed model and torch “jumps” within the tolerance range [5]. Extensions handle external axes [18], anytime refinement around a guide path minimising reconfigurations [9], dynamic programming with an explicit breakpoint cost [8], global co-optimisation of tool orientation, redundancy and timing [6] and warm-started optimisation with tolerance windows [19]. These are the deterministic baselines of the revised plan.

3.3 Task localisation and base placement

Base placement has evolved from reachability maps [20] to set-cover formulations minimising the number of stations [21], placement for sets of continuous paths [10], joint base-path optimisation [22] and placement scored by true cycle time [11]. None checks constant-speed feasibility or orientation cones at station level, which motivates stage P2 of the pipeline (Section 7.3).

3.4 Sampling-based planners guiding local optimisers

Planner-provided nominal trajectories improve the sample efficiency of MPPI [23]; MPPI can also mix samples from arbitrary ancillary controllers, including global planners [24]. This is the mechanism by which the T4.2 planner and the MPPI look-ahead are coupled.

4 Dynamic replanning and human-aware safety

4.1 Safety-aware objectives

ISO/TS 15066 speed-and-separation monitoring and power-and-force limiting reduce the robot speed near humans. Converting these limits into a time-dilation cost lets a planner minimise the *expected execution time including safety slowdowns*; on a real collaborative cell this reduced the robot time by 19% [14]. The same principle with biomechanical limits is applied in graph search [25] and in convex, jerk-limited re-timing along a fixed path [26]. These per-configuration costs are non-smooth and naturally evaluated on MPPI rollouts.

4.2 Reactive, risk-aware and constraint-aware sampling

Recent MPPI variants enforce human separation by terminating violating rollouts at a constant 22 ms per cycle on a torque-controlled arm [15], evaluate collision probabilities against non-Gaussian human predictions for crowds [27], keep a contingency plan available at all times [28], and satisfy kinematic constraints exactly by projection [29]. Torque-level MPPI provides compliant behaviour for physical interaction [30].

4.3 Implications

Safety-aware costs are evaluated in the Open IPC (MPPI rollouts, planner edges) to *anticipate* safety interventions, while the certified speed adaptation and safety rules remain in the Core IPC. Path switching and fallback strategies are provided by the multi-goal look-ahead and by the deterministic plan of each segment (Section 8). Interfaces: human tracking from Activity 3, task alternatives and allocation from Activity 2.

5 Model predictive control and MPPI for trajectory generation

This section reports the state of the art on Model Predictive Path Integral (MPPI) control and its relation to MPC, then the industrial look-ahead and path-following methods relevant for the Core-IPC micro-interpolator.

5.1 Theoretical framework

Path integral control. Consider control-affine stochastic dynamics $d\mathbf{x} = \mathbf{f}(\mathbf{x})dt + \mathbf{G}(\mathbf{x})(\mathbf{u}dt + d\mathbf{w})$, $\mathbb{E}[d\mathbf{w}d\mathbf{w}^\top] = \Sigma dt$, and the cost $J = \mathbb{E}[\phi(\mathbf{x}_T) + \int_0^T q(\mathbf{x}_t) + \frac{1}{2}\mathbf{u}_t^\top \mathbf{R}\mathbf{u}_t dt]$. The value function obeys a nonlinear Hamilton–Jacobi–Bellman (HJB) PDE. Kappen [31] showed that, under the assumption $\lambda\mathbf{R}^{-1} = \Sigma$ (noise is large where control is cheap), the exponential transformation $V = -\lambda \log \Psi$ makes the HJB *linear* in the desirability Ψ . Through the Feynman–Kac lemma the solution becomes an expectation over uncontrolled trajectories,

$$\Psi(\mathbf{x}_0, 0) = \mathbb{E}_p[\exp(-\frac{1}{\lambda}S(\tau))], \quad \mathbf{u}^*(t)dt = \frac{\mathbb{E}_p[\exp(-S(\tau)/\lambda)d\mathbf{w}_t]}{\mathbb{E}_p[\exp(-S(\tau)/\lambda)]}, \quad (1)$$

where $S(\tau) = \phi(\mathbf{x}_T) + \int q dt$ is the state cost of trajectory τ . The optimal control is thus a *cost-weighted average of noise*, computable by Monte-Carlo sampling without differentiating dynamics or cost. Theodorou *et al.* [32] exploited this result for policy improvement (PI²), and STOMP [33] for offline trajectory optimisation.

Information-theoretic MPPI. Williams *et al.* turned (1) into a receding-horizon controller [34, 35] and later re-derived it in discrete time from an information-theoretic argument that removes the control-affine assumption [2, 36]. Let $\mathbf{V} = (\mathbf{v}_0, \dots, \mathbf{v}_{T-1})$ be an input sequence sampled as $\mathbf{v}_t = \mathbf{u}_t + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$, with density q_U , and let p be a base (prior) distribution. The free energy satisfies

$$-\lambda \log \mathbb{E}_p[e^{-S(\mathbf{V})/\lambda}] \leq \mathbb{E}_q[S(\mathbf{V})] + \lambda \mathbb{D}_{\text{KL}}(q \| p), \quad (2)$$

with equality for the optimal distribution $q^*(\mathbf{V}) = \eta^{-1} \exp(-S(\mathbf{V})/\lambda) p(\mathbf{V})$. MPPI chooses the Gaussian mean $\mathbf{U}$ that minimises $\mathbb{D}_{\text{KL}}(q^* \| q_U)$, which yields a closed-form importance-sampling estimate from K rollouts:

$$\mathbf{u}_t \leftarrow \mathbf{u}_t + \sum_{k=1}^K w_k \epsilon_t^k, \quad w_k = \frac{1}{\eta} \exp\left(-\frac{1}{\lambda}(\tilde{S}_k - \rho)\right), \quad \tilde{S}_k = S(\mathbf{V}_k) + \gamma \sum_t \mathbf{u}_t^\top \Sigma^{-1} \mathbf{v}_t^k, \quad (3)$$

where $\rho = \min_k \tilde{S}_k$ ensures numerical stability, η normalises the weights, and the second term of $\tilde{S}_k$ is the likelihood-ratio (control) cost arising from the change of measure ($\gamma = \lambda(1 - \alpha)$ in [2]). The *inverse temperature* $1/\lambda$ interpolates between a plain average ($\lambda \rightarrow \infty$) and greedy selection of the best rollout ($\lambda \rightarrow 0$). Algorithm 1 summarises one control cycle; the warm-started, shifted mean makes MPPI an anytime, real-time-iteration scheme.

Properties and limitations. MPPI (i) needs neither gradients nor convexity, so collision indicators, costmaps, logical task predicates and contacts enter the cost directly; (ii) is embarrassingly parallel (GPU, and recently FPGA [37]); (iii) can use a black-box physics engine as model [38, 39]. Its known weaknesses are equally relevant for industrial use: constraints are only soft (penalties), so feasibility is not certified; sample complexity grows with input dimension and horizon; the Gaussian proposal is unimodal and the weighted *average* of rollouts belonging to different homotopy classes can be infeasible (“averaging-induced failure” [40]); injected noise produces chattering inputs [41]; and stability or recursive-feasibility guarantees of classical MPC [42] do not carry over. Most of the literature reviewed in Sect. 5.3 addresses one of these points.

5.2 MPPI versus MPC and other horizon-based planners

Gradient-based nonlinear MPC (NMPC) solves at each step a deterministic nonlinear program, typically with SQP or interior-point methods and real-time iterations (e.g. *acados* [43]); it handles hard state/input constraints, provides local optimality and, with terminal ingredients, stability and recursive feasibility [42, 44]. It is the natural tool for jerk/torque-limited reference generation, as in predictive inverse kinematics and trajectory scaling for industrial manipulators [45, 46]. Its weaknesses are the need for smooth models and costs, sensitivity to initialisation in non-convex scenes (it converges to the local minimum of the current homotopy class), and the difficulty of encoding contacts and discrete decisions. Differential dynamic programming/iLQR [47] shares these traits but adds time-varying feedback gains.

Sampling-based MPC replaces the NLP with a stochastic search. The Cross-Entropy Method [48, 49] refits a Gaussian on an elite subset, predictive sampling [39] keeps the best sample, and MPPI uses soft Boltzmann weights; Wagener *et al.* [50]

Algorithm 1 One cycle of MPPI (receding horizon)

Require: state $\mathbf{x}_0$, nominal sequence $\mathbf{U}$, covariance Σ , temperature λ , samples K , horizon T

- 1: **for** $k = 1 \dots K$ **in parallel do**
 - 2: sample $\epsilon_{0:T-1}^k \sim \mathcal{N}(\mathbf{0}, \Sigma)$; roll out $\mathbf{x}_{t+1}^k = \mathbf{F}(\mathbf{x}_t^k, \mathbf{u}_t + \epsilon_t^k)$ ▷ model or physics simulator
 - 3: $\tilde{S}_k \leftarrow \phi(\mathbf{x}_T^k) + \sum_t q(\mathbf{x}_t^k) + \gamma \mathbf{u}_t^\top \Sigma^{-1}(\mathbf{u}_t + \epsilon_t^k)$
 - 4: **end for**
 - 5: $\rho \leftarrow \min_k \tilde{S}_k$; $w_k \leftarrow \exp(-(\tilde{S}_k - \rho)/\lambda)/\eta$ ▷ importance weights
 - 6: $\mathbf{u}_t \leftarrow \mathbf{u}_t + \sum_k w_k \epsilon_t^k$; optionally smooth $\mathbf{U}$ ▷ e.g. Savitzky–Golay, splines
 - 7: apply $\mathbf{u}_0$; shift $\mathbf{U}$ by one step and re-initialise $\mathbf{u}_{T-1}$ ▷ warm start
-

unified these schemes as instances of online (mirror-descent) learning, and Stein-variational MPC [51] generalises them to non-parametric, multimodal posteriors. Classical motion-planning trajectory optimisers (CHOMP, STOMP, TrajOpt [33, 52, 53]) solve similar problems offline over full paths, while mobile-robot local planners such as DWA [54] (one-step constant-velocity sampling) and TEB [55] (sparse NLP over multiple topologies) are the industrial incumbents MPPI is replacing in Nav2 [4]. Table 1 summarises the comparison.

The emerging consensus is that the two families are complementary rather than competing: MPPI explores globally within the horizon and copes with non-smooth objectives, gradient-based MPC refines and certifies. Hybrid schemes already appear at both levels: cuRobo [17] seeds GPU L-BFGS with a particle-based (MPPI-like) optimiser and plans collision-free arm motions in about 30–50 ms, while Aoyama *et al.* [57] alternate DDP steps with Hessian-shaped sampling to escape local minima.

5.3 Analysis of the literature

5.3.1 Algorithmic advances

Robustness to disturbances and model error. Tube-MPPI [58] pairs MPPI with an iLQG ancillary tracker; RMPPI [56] adds an augmented nominal state and provides bounds on the free-energy growth, i.e. the first performance guarantee for MPPI. Ensemble MPPI [59] samples model parameters and adapts them online; U-MPPI [60] propagates means and covariances with the unscented transform and adds risk-sensitive costs; covariance steering [61] shapes the sampled trajectory distribution.

Constraints and safety. This is the most active line and the most relevant for certification. Shield-MPPI [62] and SC-MPPI [63] embed control barrier functions (CBFs) in the cost or directly in forward sampling; GS-MPPI [64] uses composite CBFs so that *all* sampled trajectories are safe; BR-MPPI [65] treats barrier rates as additional inputs and projects samples on the constraint manifold; BSS-MPPI [66] enforces chance constraints in belief space. For manipulators, PR-MPPI [29] projects sampled joint velocities on equality (closed-chain) and inequality (joint-limit, clearance) constraints and retracts the result, achieving exact constraint satisfaction on 14-DoF dual-arm systems, while COSMIK-MPPI [15] enforces human-robot separation by treating violations as rollout terminations (*constraints-as-terminations*) and reports 100% success at a constant 22 ms per cycle, where gradient-based MPC failed in cluttered scenes.

Sampling distribution, smoothness and multimodality. Smooth-MPPI [41] samples in the derivative-action space to remove chattering without post-filtering; log-MPPI [67] uses normal-log-normal noise for better exploration of costmaps; Biased-MPPI [24] re-derives MPPI for arbitrary proposal distributions and mixes samples from ancillary controllers (e.g. a global planner, a learned policy, a braking controller), alleviating local minima. MPOPI [68] adapts the proposal through adaptive importance sampling, and Tsallis-divergence VI-MPC [69] generalises the exponential weighting. Multimodality is handled by Stein variational methods (SVMPC [51], SVG-MPPI [70]), by clustering rollouts into homotopy modes before averaging (CE-MPPI [40], 48% shorter time-to-goal on a UR5e), and by learned proposals [71]. Diffusion-style annealing [72, 73] is a recent reinterpretation of the iterated path-integral update as score-based denoising.

Learning-augmented MPPI. MPPI is the planner of several model-based RL agents (POLO, MPQ, TD-MPC [74–76]) where a learned terminal value extends the effective horizon, a relevant feature for long industrial tasks with short real-time horizons.

5.3.2 Industrial and collaborative manipulators

STORM [3] is the reference joint-space MPPI for 7-DoF arms: GPU rollouts of the kinematic chain, learned self-collision and signed-distance costs, Halton-spline sampling, and a 125 Hz control rate with task- and joint-space constraints as costs. Pezzato *et al.* [38] replace the hand-coded model with Isaac Gym [77], solving navigation, non-prehensile pushing

Table 1 Qualitative comparison of horizon-based planning/control methods relevant to Plan4ARI.

	NMPC (SQP/IPM)	iLQR / DDP	CEM / predictive sampling	MPPI and variants	DWA / TEB
Optimisation principle	Deterministic NLP, Newton-type	Riccati backward pass	Elite refit / best sample	Path-integral importance sampling	Velocity sampling / sparse NLP
Gradients of model & cost	Required (2nd order)	Required	Not required	Not required	DWA: no; TEB: yes
Hard constraints	Yes (exact)	Via augmented Lagrangian	Penalties / rejection	Penalties; CBF, projection or termination in variants	Partially
Non-smooth cost, contacts, logic	Poor	Poor	Good	Good (simulator in the loop)	Limited
Multimodality	Single local optimum	Single local optimum	Unimodal Gaussian	Unimodal; multimodal in SVMPC, SVG-, M3P2I, CE-MPPI	TEB: parallel topologies
Theory	Stability, recursive feasibility	Local convergence	Weak	Free-energy optimality; tube/robust bounds [56]	Heuristic
Typical rate (arm / AMR)	100–1000 Hz (small models)	100–1000 Hz	20–200 Hz on GPU	20–170 Hz on GPU	10–50 Hz
Industrial maturity	High (acados, PLC-grade)	Medium	Research	ROS 2 Nav2, STORM, MPPI-Generic	High (ROS)

and whole-body mobile manipulation with 750 parallel simulations at 25 Hz, thus removing the need for explicit contact models. Contact-rich deployment with MuJoCo-MJX/JAX on a Franka arm [78] shows that structured multimodal sampling outperforms unimodal CEM/MPPI, but also that fidelity of contact modelling and the compute budget (8–10 Hz planning) remain bottlenecks. At the torque level, Im *et al.* [30] sample joint torques through full rigid-body dynamics, obtaining compliant, force-aware behaviour at 166 Hz with a 0.18 s horizon on a 7-DoF arm. VP-STO [79] applies the same stochastic philosophy to via-point parametrised, time-optimal trajectories, an attractive representation for industrial controllers that expect smooth, bounded references. For HRC, COSMIK-MPPI [15] closes the loop with markerless human pose estimation, which is directly aligned with the ISO/TS 15066 speed-and-separation monitoring paradigm.

5.3.3 Mobile robots and AMRs

MPPI was born on the AutoRally platform for aggressive driving [2, 34]. For indoor AMRs the key milestone is its adoption in ROS 2 Nav2, where the maintainers report that it rarely needs recovery behaviours and plan to make it the default local controller [4]. Mohamed *et al.* address navigation in unknown cluttered spaces with log-MPPI [67], Gaussian-process subgoal guidance (GP-MPPI [80]) and unscented sampling [60]. Human-aware navigation is the second major theme: DRA-MPPI [27] evaluates Monte-Carlo collision probabilities against non-Gaussian pedestrian predictions for hundreds of rollouts in real time, mitigating the freezing-robot problem; interaction-aware [81] and multi-agent path-integral formulations [82] couple the plans of several agents, a building block for fleets of AMRs. Contingency-MPPI [28] guarantees that a safe fallback plan always exists, and the FPGA design of [37] reduces latency ($3.1\text{--}7.5\times$) and energy ($2.5\text{--}5.4\times$) on battery-powered embedded platforms.

5.3.4 In-depth: Multi-Modal MPPI and Active Inference for reactive TAMP

Zhang *et al.* [83] address *reactive* task-and-motion planning, the closest problem to Plan4ARI’s mobile-manipulation scenarios. A high-level Active Inference planner (AIP) [84], guided by a Behaviour Tree, reasons on symbolic beliefs and is modified to return not one but N *alternative* plans (e.g. push vs. pull, top vs. side grasp); each first action is linked through a plan interface to a cost function C_i . The low-level Multi-Modal MPPI (M3P2I) then samples $N \cdot K$ control sequences in Isaac Gym, K per alternative, with Halton-spline noise. Weights are first computed *per alternative*, $\omega_i(\mathbf{V}_k) \propto \exp(-(S_i(\mathbf{V}_k) - \rho_i)/\beta_i)$, with the inverse temperature β_i auto-tuned so that the effective sample count η_i

stays in 5–10% of K ; the per-mode means μ_i are then concatenated and re-weighted jointly to produce a single command $u_t = (1 - \alpha_u)u_{t-1} + \alpha_u \sum_k \tilde{\omega}_k v_{t,k}$. The action planner runs at about 1 Hz and M3P2I at 25 Hz.

Results. In a push–pull task with an omnidirectional base, the multimodal controller always succeeded, whereas push-only timed out when the object lay in a corner; mean orientation error dropped from 0.0198 (push) and 0.0777 (pull) to 0.0041 and execution time from 6.2–25.1 s to 3.8 s (object in the middle). In a 7-DoF pick-and-place task with disturbances (moved/stolen cube, dynamic obstacles) it outperformed an Isaac Gym actor-critic RL baseline (position error 0.0117 vs. 0.0246 m) without any training, and it blended top and side grasps according to the geometry; real experiments on a Franka Panda confirmed recovery from disturbances.

Assessment for Plan4ARI. The key idea, moving the choice between discrete alternatives from hand-written task-level heuristics to parallel sampling at the motion level, is directly reusable for alternative grasps, approach directions, base placements or homotopy classes produced by a global planner. Its limitations are equally instructive: (i) the final weighted blending can still average incompatible modes (the issue later formalised by CE-MPPI [40]); (ii) all constraints, including obstacle avoidance, are soft costs with manually tuned weights; (iii) computation scales with $N \cdot K$ full physics rollouts, and fidelity depends on the simulator (sim-to-real handled only by randomisation); (iv) no explicit treatment of humans or safety standards. These gaps are complementary to those solved by the constraint-aware and risk-aware variants above.

5.3.5 Software ecosystem

Mature open-source implementations exist: MPPI-Generic (CUDA, C++) [85], the Nav2 MPPI controller [4], STORM and cuRobo for arms [3, 17], Isaac Gym-based MPPI [38], MuJoCo predictive sampling [39] and JAX/MJX pipelines [78]. None of them, however, targets the deterministic timing and safety-rated interfaces of industrial controllers.

5.4 Industrial look-ahead, path following and micro-interpolation

Industrial controllers adapt the speed along a buffered path under kinematic limits. A path-accurate online trajectory generator modifies only the time law of a sampled joint path under velocity, acceleration and jerk limits at 250 Hz on an industrial controller [7]; CNC-style Cartesian look-ahead blends corners and scans speed limits bidirectionally [86]. Model predictive path-following control treats the path parameter as a virtual state and was solved in at most 0.48 ms at 1 kHz on an industrial arm [87]. Predictive inverse kinematics with task scaling and predictive joint trajectory scaling [45, 46] and null-space NMPC [88] are the deterministic counterparts of the MPPI look-ahead. Together they define the path–velocity layer of the Core IPC (Section 7): the joint path is fixed upstream, the time law is optimised within the process speed band under hard limits, and band violations are reported upstream instead of producing unplanned stops.

6 Software frameworks and industrial toolchain

Table 2 Framework matrix (candidates for the Plan4ARI toolchain).

Tool	Language / hardware	Scope	Role in Plan4ARI
OMPL / MoveIt 2	C++, CPU	sampling-based planners, ROS 2 integration	KPI baseline (proposal)
VAMP [1]	C++/Python, CPU SIMD	vectorised sampling-based planning	transfers, station checks, rollout primitives
Descartes [5]	C++, ROS	tolerance-aware Cartesian path planning	deterministic baseline
cuRobo [17]	CUDA	GPU motion generation	GPU track baseline
MPPI-Generic [85]	CUDA C++	MPPI, Tube-MPPI, RMPPI	GPU MPPI reference
STORM [3]	PyTorch, GPU	joint-space MPPI for arms	MPPI reference
Nav2 MPPI controller [4]	C++, CPU	AMR local control	AMR baseline
acados [43]	C, code generation	embedded NMPC	Core-IPC MPC candidate
Isaac Gym / MuJoCo-MJX [77, 78]	GPU physics	simulator-in-the-loop sampling MPC	simulation benchmarks
MI.RA/Dexter (COMAU)	dockerised actions	low-code platform, digital twin	integration target (T4.4)

Logging, benchmarking, continuous integration and traceability are handled in T4.6 and in the benchmark repository (Section 9).

7 Gap analysis and revised architecture

7.1 What changes with respect to the proposal

Table 3 Original plan (Project Description, §A.4 and §E.3) versus updated plan.

	Original plan	Updated plan
Global / online planning (T4.2)	Informed sampling-based planner, mixed global/local sampling, multi-path reuse	Retained ; also used for transfer motions (retract, reconfigure, approach) and station-level feasibility; vectorised CPU primitives [1]
Local reactive layer	MPC “alongside” the sampling-based planner	New: MPPI look-ahead over the task null space with multiple goals, guided by the sampling-based planner [23, 24]
Micro-interpolation (T4.3)	MPC in the Core IPC, speed override, visual servoing	Retained ; specified as path-velocity layer with hard limits and speed band [7, 46, 87]
Task localisation	Implicit (industrial practice)	Explicit pipeline stage : optimal placement of robot / AGV / external axis [10, 21]
Safety-aware cost	ISO/TS 15066 cost in the replanner [14]	Retained ; evaluated per rollout and per planner edge
Robot dependence	COMAU platform	Robot-agnostic kinematic abstraction
Benchmarking (T4.7)	≥ 50 scenarios, OMPL baseline, GPU planners not compared	Same KPIs; application catalogue; welding on hardware, others in simulation; GPU methods in a separate track

7.2 Updated approach: sampling-based planning and MPPI

Industrial controllers rely on look-ahead: a buffer of future path samples is analysed to adapt speed under kinematic limits [7]. For under-constrained tasks this look-ahead is limited to a single, pre-computed joint solution. MPPI acts as a *stochastic look-ahead* that explores the redundancy ahead of the robot: it samples many short futures of the null-space variables (torch roll, cone tilt, external axis), evaluates them with arbitrary, non-smooth costs (collisions with the structure, joint-limit and singularity margins, ISO/TS 15066 slowdown, speed feasibility) and returns an improved plan every cycle [2]. Null-space sampling with exact task satisfaction [29, 89], its deterministic counterpart [88] and planner-guided MPPI [23, 24] support this choice. Sampling-based planning remains essential where MPPI is weak: long transfer motions in clutter, station-level feasibility and multi-path reuse. *The sampling-based planner proposes, MPPI refines and reacts, the Core-IPC MPC certifies.*

7.2.1 Multi-goal MPPI look-ahead

The goal of the look-ahead is a Cartesian task segment (e.g. ~ 10 cm of seam) with a *set of start configurations*, one per inverse-kinematics solution class (up to 8 for a non-cuspidal 6R arm, up to 16 for cuspidal arms, multiplied by joint turns where ranges exceed $\pm\pi$ [16]). For each class MPPI optimises the null-space variables along the segment with exact IK on that class. Classes are *selected, never blended* [40, 90]. Continuing on the current class has no transfer cost; switching class costs a transfer (planned by the sampling-based planner first to the nominal start for a cost estimate, then to the optimised start, with a consistency check) plus the restart overlap. The time-dilation factor $\lambda = \max_j |q_j^{\text{req}}|/q_j^{\text{max}}$ along each rollout measures speed feasibility ($\lambda \leq 1/0.9$ for a $\pm 10\%$ band), so that a reconfiguration emerges from cost comparison when the current class cannot keep the speed or approaches joint limits.

7.3 End-to-end pipeline

Po – Scene and task model. CAD of workpiece and cell registered with the sensing modules of Activity 3; local registration at each station (R5).

P1 – Task segmentation. Segments with homogeneous constraints (welding: seams split at corners, interruptions and process changes; corners at constant speed are where joint-speed limits break [91]).

P2 – Optimal task localisation. Minimum set of stations (AGV poses, rail intervals or fixed base) such that each segment is *path-wise feasible* from at least one station: set-cover over segments [21] with a continuous-branch test [10], the speed

stay comparable with the OMPL-based KPI; GPU execution is a separately reported variant. (iii) Every online plan has a deterministic fallback: the P3 plan of the current segment. (iv) Fixed-seed, low-discrepancy sampling and logging make execution repeatable for process qualification.

7.5 Robot-agnostic design

The framework is parametrised by a *kinematic description* (URDF, joint ranges, external axes, base model) and a *task description* (constraint type per Cartesian degree of freedom: exact, band, cone, free). From these it derives automatically: the IK solver class — analytic where available (6R families, including cuspidal arms [16]), numerical functional-redundancy IK otherwise [93] — and the set of *solution classes* used as goals (branches $\times$ joint turns); the null-space coordinates sampled by MPPI (tool roll, cone tilts, external axes, redundant joints, base); collision geometry (sphere approximations for vectorised checking) and kinematic limits. Supported configurations: fixed arms, arms on linear or rotary axes, arms on AGVs (base fixed per station or interpolated), dual-arm and multi-robot cells [94].

8 Selected methodologies and fallback options

Table 5 Method-selection record.

Function	Selected approach	Baselines	Fallback
Task localisation (P2)	Set-cover over segments with path-wise feasibility test	Point-based placement [20, 21]; optimisation-based [10, 22]	Operator-defined stations
Segment feasibility, cost-to-go (P3)	Layered graph / DP over IK classes and discretised null space, breakpoint cost	Descartes [5]; anytime tracking [9]; co-optimisation [6]	Single-class DP
Transfer and global planning (T4.2)	Informed sampling with mixed global/local sampling and multi-path reuse, vectorised primitives	OMPL planners; VAMP [1]	Pre-taught via-points
Online look-ahead (P4)	Multi-goal null-space MPPI, exact IK per class, select-never-blend, planner-guided	Null-space NMPC [88]; predictive IK [45]; P3 plan replay	P3 plan + speed scaling
Micro-interpolation (P5, T4.3)	Path-velocity MPC with speed band and hard limits	Path-accurate OTG [7]; trajectory scaling [26, 46]	Speed override only
Human-aware cost	ISO/TS 15066 time-dilation cost [14]; risk-aware rollouts [27]	Plain SSM speed scaling	Safety stop

Verification obligations. Exact task tracking and hard limits are guaranteed by construction (exact IK per class, Core-IPC MPC); MPPI provides performance, not certification. Optimality and completeness claims are attached to the deterministic layers (P3, T4.2 planner). Fallbacks cover infeasible optimisation (P3 plan), missed deadlines (speed scaling in the Core IPC) and degraded sensing (stop at the next planned detach point).

9 Benchmarking and evaluation plan (Task 4.7)

The methods are evaluated on a broad catalogue of industrial applications sharing the structure “under-constrained task + speed requirements + possible base/axis redundancy + dynamic, human-shared scene”. **Arc welding of large structures is the only application validated on real hardware** (reference demonstrator); **all other applications are benchmarked in simulation**, on digital twins (COMAU virtualisation tools and open simulators), across several robot kinematic classes (R7).

9.1 Application catalogue

Table 6 Application catalogue for Task 4.7 (R = real hardware, S = simulation).

Application	Task structure / redundancy	Platform	Key metrics	Eval.
Arc welding of large structures	5-DoF seam, cone, free roll, constant speed $\pm 10\%$, reconfigurations	6-axis arm on AGV, optional rail	restarts, speed band, angles	R + S
Multi-robot welding	synchronous / ordered seams, shared workspace [92]	2+ arms, AGVs	cycle time, waiting, collisions	S
Spray painting / coating	5-DoF, free roll about nozzle, stand-off band [19]	arm, rail	coverage, speed band	S
Cold / thermal spray	5-DoF toolpath, functional redundancy [93]	arm	joint travel, speed	S
Sealing / gluing / dispensing	5-DoF, constant speed, corners	arm, AGV	speed band, path error	S
Deburring / grinding / sanding	5-DoF, contact force, large parts [12]	mobile manipulator	path error, force, stations	S
Milling / trimming / machining	5-DoF, stiffness, smooth tool axis [95, 96]	arm, positioner, dual arm [94]	error, jerk, cycle time	S
Polishing	5-DoF, contact, cone	arm	speed, force	S
Laser cutting / laser welding	5-DoF, tight speed tolerance	arm, gantry	speed band, error	S
Drilling / riveting (aerospace)	point targets with tool axis, many stations [21]	mobile manipulator	stations, cycle time	S
Ultrasound / NDT inspection	surface following with bounded offsets [97]	arm, AGV	coverage, offsets	S
Wire-arc additive manufacturing	5-DoF layers, free roll [98]	arm, positioner	speed, joint travel	S
Fibre / tape placement	tool normal to surface, redundant cell [99]	arm, positioner, axes	singularity margin, time	S
Pick-and-place / palletising	free-space transfers, base placement [11]	arm, AGV	cycle time, success	S
Machine tending / assembly	transfers in clutter, human access	cobot, AGV	success, latency, idle time	S
Collaborative operations near humans	ISO/TS 15066 SSM/PFL [14, 15]	cobot	idle time (-30% target), safety stops	S
Intralogistics / AMR fleets	navigation in crowds [4, 27]	AMRs, mobile manipulators	success, freezing, time	S

9.2 Protocol and KPIs

In line with the proposal: at least 50 application scenarios drawn from Table 6, computation time averaged over more than 100 contexts, OMPL planners as baseline for the sampling-based layer; GPU-based variants reported in a separate

track. Additional baselines per layer as in Table 5. Metrics: success rate; planning and replanning latency; number of stations, restarts and reconfigurations; deviation from the speed band; task-tolerance usage (work/travel angles); joint-limit and singularity margins; path accuracy; cycle time; safety-induced idle time and safety stops; smoothness (jerk); CPU/memory load; repeatability across runs. The welding benchmark extends an existing welding motion-planning benchmark framework (weld segments plus transitions) [100]. Ablations: with/without MPPI look-ahead, with/without P2 optimisation, horizon length (5/10/20 cm), number of goals, CPU vs GPU.

9.3 Benchmark repository

Scenarios, robot models, task descriptions, baselines and evaluation scripts will be collected in a version-controlled repository (to be set up within T4.6/T4.7), so that results are reproducible and comparable across partners.

10 Integration roadmap, risks and compliance

10.1 Mapping onto Tasks 4.2–4.7

- **T4.2** (M4–M24): informed sampling-based planner with mixed sampling and multi-path reuse (as proposed); in addition, vectorised CPU primitives, transfer motions for reconfiguration, station-level feasibility (P2–P3) and the MPPI look-ahead (P4), including its guidance by the sampling-based planner.
- **T4.3** (M4–M24): path-velocity MPC micro-interpolator (P5) with speed band, hard limits, Cartesian/joint interfaces and synchronous sensor input (as proposed); interface to the MPPI layer.
- **T4.4** (M25–M30): integration of P1–P4 as dockerised MI.RA/Dexter actions.
- **T4.5** (M31–M36): integration of P5 in the COMAU Core IPC.
- **T4.6** (M4–M36): continuous integration, including the benchmark repository.
- **T4.7** (M1–M6, then continuous): application catalogue, scenarios and protocol (Section 9).

Interfaces and dependencies: Activity 2 (task alternatives, multi-robot allocation), Activity 3 (registration, human tracking, seam sensing), Activity 5 (motion/neural control in the Core IPC), Activity 7 (demonstrator).

10.2 Risks

Table 7 Main technical risks and mitigations.

Risk	Mitigation
Novelty boundary: null-space sampling MPPI alone is not new [89]	Claim the combination of online null-space look-ahead with solution-class modes, planned transfers and restarts, speed band, task localisation and Open/Core IPC deployment
Compute budget of MPPI on industrial CPUs	Early micro-benchmark in T4.2; CPU SIMD primitives; GPU as optional track
Accuracy of mobile platforms on large parts [12]	Local registration and seam tracking (Activity 3) as prerequisite
Process coupling (restart overlap, speed tolerance)	Parameters agreed with welding process experts; parametric in the planner
KPI comparability (GPU planners excluded from the OMPL comparison)	CPU-first design, separate GPU track
Certification of stochastic planners	Hard limits and safety enforced only in the deterministic Core IPC; repeatable sampling and logging

10.3 Standards and compliance

ISO 10218 and ISO/TS 15066 for collaborative operation (speed and separation monitoring, power and force limiting); welding quality requirements (process qualification, repeatability of the executed trajectory); EU AI Act requirements on transparency and safety of AI components, as foreseen by the Project Description.

11 Conclusions and approved updates

11.1 Retained and modified methods

- Retained: informed sampling-based planning with mixed sampling and multi-path reuse (T4.2); MPC micro-interpolation in the Core IPC (T4.3); ISO/TS 15066 safety-aware cost.
- Added: MPPI look-ahead with multiple goals over the task null space; explicit optimal task localisation (stations, base, rail); deterministic segment planner providing cost-to-go and fallbacks; robot-agnostic kinematic abstraction.
- Modified: the Core-IPC MPC is specified as a path-velocity layer with process speed band; the sampling-based planner is also used for transfer motions and station-level checks.

11.2 Approved benchmark and integration plan

Application catalogue of Section 9, with welding on real hardware and all other applications in simulation; KPIs and OMPL baseline as in the proposal, GPU variants in a separate track; benchmark repository within T4.6/T4.7.

11.3 Open scientific questions and technology watch

Horizon length versus number of restarts for the look-ahead; mode selection stability (hysteresis) among solution classes; certification argument for the MPPI + Core-IPC chain; scaling to tightly coupled multi-robot tasks; updates of fast sampling-based planners and of MPPI theory (gradient and expectation-maximisation interpretations [90, 101]).

11.4 Consortium approval record

To be completed after the consortium review (date, participants, decisions).
--

References

- [1] W. Thomason, Z. Kingston, and L. E. Kavraki, "Motions in microseconds via vectorized sampling-based planning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 8749–8756, arXiv:2309.14545.
- [2] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, "Information-theoretic model predictive control: Theory and applications to autonomous driving," *IEEE Transactions on Robotics*, vol. 34, no. 6, pp. 1603–1622, 2018, arXiv:1707.02342.
- [3] M. Bhardwaj, B. Sundaralingam, A. Mousavian, N. Ratliff, D. Fox, F. Ramos, and B. Boots, "STORM: An integrated framework for fast joint-space model-predictive control for reactive manipulation," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 164, 2022, pp. 750–759, arXiv:2104.13542.
- [4] S. Macenski, T. Moore, D. Lu, A. Merzlyakov, and M. Ferguson, "From the desks of ROS maintainers: A survey of modern & capable mobile robotics algorithms in the robot operating system 2," *Robotics and Autonomous Systems*, vol. 168, p. 104493, 2023, arXiv:2307.15236.
- [5] J. De Maeyer, B. Moyaers, and E. Demeester, "Cartesian path planning for arc welding robots: Evaluation of the descartes algorithm," in *2017 22nd IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2017, pp. 1–8.
- [6] Y. Chen, T. Zhang, Y. Huang, T. Liu, and C. C. L. Wang, "Co-optimization of tool orientations, kinematic redundancy, and waypoint timing for robot-assisted manufacturing," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 12 102–12 117, 2025, arXiv:2409.13448.
- [7] F. Lange and A. Albu-Schaffer, "Path-accurate online trajectory generation for jerk-limited industrial robots," *IEEE Robotics and Automation Letters*, vol. 1, no. 1, pp. 82–89, 2016.
- [8] Z. Yin, F. Wu, R. Bian, Z. Wang, J. Yang, J. Tan, and D. Kong, "Dynamic programming-based redundancy resolution for path planning of redundant manipulators considering breakpoints," arXiv preprint arXiv:2411.17034, 2024. [Online]. Available: <https://arxiv.org/abs/2411.17034>
- [9] Y. Wang and M. Gleicher, "Anytime planning for end-effector trajectory tracking," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3246–3253, 2025, arXiv:2502.03676.
- [10] T. Weingartshofer, C. Hartl-Nesic, and A. Kugi, "Optimal TCP and robot base placement for a set of complex continuous paths," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 9659–9665.
- [11] A. Wachter, A. Kugi, and C. Hartl-Nesic, "Time-optimal TCP and robot base placement for pick-and-place tasks in highly constrained environments," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 2251–2257.
- [12] X. Zhao, B. Tao, S. Han, and H. Ding, "Accuracy analysis in mobile robot machining of large-scale workpiece," *Robotics and Computer-Integrated Manufacturing*, vol. 71, p. 102153, 2021.
- [13] H. Wang, X. Wang, C. Liu, C. Chen, and X. Zhao, "Innovation in robot welding trajectory control: Online synchronous planning of welding paths and torch postures for structures with unknown spatial geometries," *Robotics and Computer-Integrated Manufacturing*, vol. 101, p. 103315, 2026.
- [14] M. Faroni, M. Beschi, and N. Pedrocchi, "Safety-aware time-optimal motion planning with uncertain human state estimation," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 12 219–12 226, 2022, arXiv:2210.11655.
- [15] E. Gursay, M. Sabbah, A. Haffemayer, J. Cavalcanti Santos, P. N. Crestaz, V. Petrik, N. Mansard, and V. Bonnet, "COSMIK-MPPI: Scaling constrained model predictive control to collision avoidance in close-proximity dynamic human environments," arXiv preprint arXiv:2604.10358, 2026. [Online]. Available: <https://arxiv.org/abs/2604.10358>
- [16] A. J. Elias and J. T. Wen, "Path planning and optimization for cuspidal 6R manipulators," *Journal of Mechanisms and Robotics*, vol. 17, no. 12, p. 121008, 2025, arXiv:2501.18505.
- [17] B. Sundaralingam, S. K. S. Hari, A. Fishman, C. Garrett, K. V. Wyk, V. Blukis, A. Millane, H. Oleynikova, A. Handa, F. Ramos, N. Ratliff, and D. Fox, "cuRobo: Parallelized collision-free robot motion generation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 8112–8119, arXiv:2310.17274.

- [18] Y. Sun, X. Meng, M. Yu, and H. Tang, "Semi-constrained path planning for industrial robot with external axis," in *2020 Chinese Automation Congress (CAC)*, 2020, pp. 6180–6185.
- [19] T. Weingartshofer, B. Bischof, M. Meiringer, C. Hartl-Nesic, and A. Kugi, "Optimization-based path planning framework for industrial manufacturing processes with complex continuous paths," *Robotics and Computer-Integrated Manufacturing*, vol. 82, p. 102516, 2023.
- [20] A. Makhal and A. K. Goins, "Reuleaux: Robot base placement by reachability analysis," in *Proc. IEEE Int. Conf. Robotic Computing (IRC)*, 2018, pp. 137–142, arXiv:1710.01328.
- [21] Q.-N. Nguyen, N. Adrian, and Q.-C. Pham, "Task-space clustering for mobile manipulator task sequencing," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 3693–3699, arXiv:2305.17345.
- [22] Z. Zhao, L. Cui, S. Xie, S. Zhang, Z. Han, L. Ruan, and Y. Zhu, "B*: Efficient and optimal base placement for fixed-base manipulators," *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10 634–10 641, 2025, arXiv:2504.12719.
- [23] C. Tao, H. Kim, and N. Hovakimyan, "RRT guided model predictive path integral method," in *2023 American Control Conference (ACC)*, 2023, pp. 776–781, arXiv:2301.13143.
- [24] E. Trevisan and J. Alonso-Mora, "Biased-MPPI: Informing sampling-based model predictive control by fusing ancillary controllers," *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5871–5878, 2024, arXiv:2401.09241.
- [25] R. Laha, W. Wu, R. Sun, N. Mansfeld, L. F. C. Figueredo, and S. Haddadin, "S*: On safe and time efficient robot motion planning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 12 758–12 764.
- [26] A. Palleschi, M. Hamad, S. Abdolshah, M. Garabini, S. Haddadin, and L. Pallottino, "Fast and safe trajectory planning: Solving the cobot performance/safety trade-off in human-robot shared environments," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5445–5452, 2021.
- [27] E. Trevisan, K. A. Mustafa, G. Notten, X. Wang, and J. Alonso-Mora, "Dynamic risk-aware MPPI for mobile robots in crowds via efficient Monte Carlo approximations," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2025, pp. 313–320, arXiv:2506.21205.
- [28] L. Jung, A. Estornell, and M. Everett, "Contingency constrained planning with MPPI within MPPI," arXiv preprint arXiv:2412.09777, 2024. [Online]. Available: <https://arxiv.org/abs/2412.09777>
- [29] S. Lee, L. Park, M. Kang, and S. Kim, "Projection-retraction MPPI: Exact constraint-manifold control for manipulators," arXiv preprint arXiv:2608.07573, 2026. [Online]. Available: <https://arxiv.org/abs/2608.07573>
- [30] E. Im, T. Kim, Y. Oh, M. Lim, and Y. Lee, "Real-time dynamics-based torque-sampling MPPI for compliant and force aware manipulation," arXiv preprint arXiv:2609.02020, 2026. [Online]. Available: <https://arxiv.org/abs/2609.02020>
- [31] H. J. Kappen, "Path integrals and symmetry breaking for optimal control theory," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2005, no. 11, p. P11011, 2005, arXiv:physics/0505066.
- [32] E. Theodorou, J. Buchli, and S. Schaal, "A generalized path integral control approach to reinforcement learning," *Journal of Machine Learning Research*, vol. 11, pp. 3137–3181, 2010.
- [33] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, "STOMP: Stochastic trajectory optimization for motion planning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2011, pp. 4569–4574.
- [34] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, "Aggressive driving with model predictive path integral control," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2016, pp. 1433–1440.
- [35] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.
- [36] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou, "Information theoretic MPC for model-based reinforcement learning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2017, pp. 1714–1721.
- [37] T. Desai, B. Plancher, and R. I. Bahar, "Real-time, energy-efficient, sampling-based optimal control via FPGA acceleration," arXiv preprint arXiv:2601.17231, 2026. [Online]. Available: <https://arxiv.org/abs/2601.17231>

- [38] C. Pezzato, C. Salmi, E. Trevisan, M. Spahn, J. Alonso-Mora, and C. Hernández Corbato, "Sampling-based model predictive control leveraging parallelizable physics simulations," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2750–2757, 2025, arXiv:2307.09105.
- [39] T. Howell, N. Gileadi, S. Tunyasuvunakool, K. Zakka, T. Erez, and Y. Tassa, "Predictive sampling: Real-time behaviour synthesis with MuJoCo," arXiv preprint arXiv:2212.00541, 2022. [Online]. Available: <https://arxiv.org/abs/2212.00541>
- [40] Z. Liu, K. Chang, and X. Chen, "Clustering-embedded model predictive path integral control: Avoiding averaging-induced failure and enabling efficient cluster selection for dynamic obstacles," arXiv preprint arXiv:2607.06499, 2026. [Online]. Available: <https://arxiv.org/abs/2607.06499>
- [41] T. Kim, G. Park, K. Kwak, J. Bae, and W. Lee, "Smooth model predictive path integral control without smoothing," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10 406–10 413, 2022, arXiv:2112.09988.
- [42] D. Q. Mayne, "Model predictive control: Recent developments and future promise," *Automatica*, vol. 50, no. 12, pp. 2967–2986, 2014.
- [43] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [44] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Madison, WI: Nob Hill Publishing, 2017.
- [45] M. Faroni, M. Beschi, N. Pedrocchi, and A. Visioli, "Predictive inverse kinematics for redundant manipulators with task scaling and kinematic constraints," *IEEE Transactions on Robotics*, vol. 35, no. 1, pp. 278–285, 2019.
- [46] M. Faroni, M. Beschi, C. Guarino Lo Bianco, and A. Visioli, "Predictive joint trajectory scaling for manipulators with kinodynamic constraints," *Control Engineering Practice*, vol. 95, p. 104264, 2020.
- [47] Y. Tassa, T. Erez, and E. Todorov, "Synthesis and stabilization of complex behaviors through online trajectory optimization," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 4906–4913.
- [48] R. Rubinstein, "The cross-entropy method for combinatorial and continuous optimization," *Methodology and Computing in Applied Probability*, vol. 1, no. 2, pp. 127–190, 1999.
- [49] C. Pinneri, S. Sawant, S. Blaes, J. Achterhold, J. Stueckler, M. Rolinek, and G. Martius, "Sample-efficient cross-entropy method for real-time planning," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 155, 2020, arXiv:2008.06389.
- [50] N. Wagener, C.-A. Cheng, J. Sacks, and B. Boots, "An online learning approach to model predictive control," in *Proc. Robotics: Science and Systems (RSS) XV*, 2019, arXiv:1902.08967.
- [51] A. Lambert, A. Fishman, D. Fox, B. Boots, and F. Ramos, "Stein variational model predictive control," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 155, 2020, arXiv:2011.07641.
- [52] M. Zucker, N. Ratliff, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, C. M. Dellin, J. A. Bagnell, and S. S. Srinivasa, "CHOMP: Covariant Hamiltonian optimization for motion planning," *The International Journal of Robotics Research*, vol. 32, no. 9–10, pp. 1164–1193, 2013.
- [53] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, "Motion planning with sequential convex optimization and convex collision checking," *The International Journal of Robotics Research*, vol. 33, no. 9, pp. 1251–1270, 2014.
- [54] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [55] C. Rösmann, F. Hoffmann, and T. Bertram, "Integrated online trajectory planning and optimization in distinctive topologies," *Robotics and Autonomous Systems*, vol. 88, pp. 142–153, 2017.
- [56] M. Gandhi, B. Vlahov, J. Gibson, G. Williams, and E. A. Theodorou, "Robust model predictive path integral control: Analysis and performance guarantees," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1423–1430, 2021, arXiv:2102.09027.

- [57] Y. Aoyama, M. Jung, A. Ratheesh, and E. A. Theodorou, "Beyond pure sampling: Hybrid optimization mechanisms for non-convex model predictive control," arXiv preprint arXiv:2606.00737, 2026. [Online]. Available: <https://arxiv.org/abs/2606.00737>
- [58] G. Williams, B. Goldfain, P. Drews, K. Saigol, J. M. Rehg, and E. A. Theodorou, "Robust sampling based model predictive control with sparse objective information," in *Proc. Robotics: Science and Systems (RSS) XIV*, 2018.
- [59] I. Abraham, A. Handa, N. Ratliff, K. Lowrey, T. D. Murphey, and D. Fox, "Model-based generalization under parameter uncertainty using path integral control," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 2864–2871, 2020, arXiv:2006.03106.
- [60] I. S. Mohamed, J. Xu, G. S. Sukhatme, and L. Liu, "Towards efficient MPPI trajectory generation with unscented guidance: U-MPPI control strategy," *IEEE Transactions on Robotics*, vol. 41, pp. 1172–1192, 2025, arXiv:2306.12369.
- [61] J. Yin, Z. Zhang, E. Theodorou, and P. Tsotras, "Trajectory distribution control for model predictive path integral control using covariance steering," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2022, pp. 1478–1484, arXiv:2109.12147.
- [62] J. Yin, C. Dawson, C. Fan, and P. Tsotras, "Shield model predictive path integral: A computationally efficient robust MPC approach using control barrier functions," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7106–7113, 2023, arXiv:2302.11719.
- [63] M. Gandhi, H. Almubarak, and E. Theodorou, "Safe importance sampling in model predictive path integral control," arXiv preprint arXiv:2303.03441, 2023. [Online]. Available: <https://arxiv.org/abs/2303.03441>
- [64] P. Rabiee and J. B. Hoagg, "Guaranteed-safe MPPI through composite control barrier functions for efficient sampling in multi-constrained robotic systems," in *Proc. IEEE Conf. Decision and Control (CDC)*, 2025, pp. 5515–5520, arXiv:2410.02154.
- [65] H. Parwana, T. Kim, K. Long, B. Hoxha, H. Okamoto, G. Fainekos, and D. Panagou, "BR-MPPI: Barrier-rate guided MPPI for enforcing multiple inequality constraints with learned signed distance fields," arXiv preprint arXiv:2506.07325, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07325>
- [66] J. Yin, P. Tsotras, and K. Berntorp, "Chance-constrained information-theoretic stochastic model predictive control with safety shielding," in *Proc. IEEE Conf. Decision and Control (CDC)*, 2024, pp. 653–658, arXiv:2408.00494.
- [67] I. S. Mohamed, K. Yin, and L. Liu, "Autonomous navigation of AGVs in unknown cluttered environments: log-MPPI control strategy," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10 240–10 247, 2022, arXiv:2203.16599.
- [68] D. M. Asmar, R. Senanayake, S. Manuel, and M. J. Kochenderfer, "Model predictive optimized path integral strategies," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 3182–3188, arXiv:2203.16633.
- [69] Z. Wang, O. So, J. Gibson, B. Vlahov, M. S. Gandhi, G.-H. Liu, and E. A. Theodorou, "Variational inference MPC using Tsallis divergence," in *Proc. Robotics: Science and Systems (RSS) XVII*, 2021, arXiv:2104.00241.
- [70] K. Honda, N. Akai, K. Suzuki, M. Aoki, H. Hosogaya, H. Okuda, and T. Suzuki, "Stein variational guided model predictive path integral control: Proposal and experiments with fast maneuvering vehicles," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 7020–7026, arXiv:2309.11040.
- [71] J. Sacks and B. Boots, "Learning sampling distributions for model predictive control," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 205, 2023, arXiv:2212.02587.
- [72] H. Xue, C. Pan, Z. Yi, G. Qu, and G. Shi, "Full-order sampling-based MPC for torque-level locomotion control via diffusion-style annealing," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2025, pp. 4974–4981, arXiv:2409.15610.
- [73] C. Pan, Z. Yi, G. Shi, and G. Qu, "Model-based diffusion for trajectory optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 57 914–57 943, arXiv:2407.01573.
- [74] K. Lowrey, A. Rajeswaran, S. Kakade, E. Todorov, and I. Mordatch, "Plan online, learn offline: Efficient learning and exploration via model-based control," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019, arXiv:1811.01848.
- [75] M. Bhardwaj, A. Handa, D. Fox, and B. Boots, "Information theoretic model predictive q-learning," in *Proc. Learning for Dynamics and Control (L4DC)*, PMLR vol. 120, 2020, arXiv:2001.02153.

- [76] N. Hansen, X. Wang, and H. Su, “Temporal difference learning for model predictive control,” in *Proc. Int. Conf. Machine Learning (ICML)*, PMLR vol. 162, 2022, arXiv:2203.04955.
- [77] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, “Isaac gym: High performance GPU-Based physics simulation for robot learning,” arXiv preprint arXiv:2108.10470, 2021. [Online]. Available: <https://arxiv.org/abs/2108.10470>
- [78] M. Dierking, J. Carvalho, A. T. Le, G. Chalvatzaki, and J. Peters, “Real-world deployment of massively parallel sampling-based MPC for contact-rich manipulation,” arXiv preprint arXiv:2606.20712, 2026. [Online]. Available: <https://arxiv.org/abs/2606.20712>
- [79] J. Jankowski, L. Brudermüller, N. Hawes, and S. Calinon, “VP-STO: Via-point-based stochastic trajectory optimization for reactive robot behavior,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 10 125–10 131, arXiv:2210.04067.
- [80] I. S. Mohamed, M. Ali, and L. Liu, “GP-guided MPPI for efficient navigation in complex unknown cluttered environments,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2023, pp. 7463–7470, arXiv:2307.04019.
- [81] W. Jansma, E. Trevisan, Á. Serra-Gómez, and J. Alonso-Mora, “Interaction-aware sampling-based MPC with learned local goal predictions,” in *Proc. Int. Symp. Multi-Robot and Multi-Agent Systems (MRS)*, 2023, pp. 15–21, arXiv:2309.14931.
- [82] L. Streichenberg, E. Trevisan, J. J. Chung, R. Siegwart, and J. Alonso-Mora, “Multi-agent path integral control for interaction-aware motion planning in urban canals,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 1379–1385, arXiv:2302.06547.
- [83] Y. Zhang, C. Pezzato, E. Trevisan, C. Salmi, C. Hernández Corbato, and J. Alonso-Mora, “Multi-modal MPPI and active inference for reactive task and motion planning,” *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7461–7468, 2024, arXiv:2312.02328.
- [84] C. Pezzato, C. Hernández Corbato, S. Bonhof, and M. Wisse, “Active inference and behavior trees for reactive action planning and execution in robotics,” *IEEE Transactions on Robotics*, vol. 39, no. 2, pp. 1050–1069, 2023.
- [85] B. Vlahov, J. Gibson, M. Gandhi, and E. A. Theodorou, “MPPI-Generic: A CUDA library for stochastic trajectory optimization,” arXiv preprint arXiv:2409.07563, 2024. [Online]. Available: <https://arxiv.org/abs/2409.07563>
- [86] J. Ma, H. Tang, R. Guo, C. Wu, U. A. Bhatti, and C. W. Tong, “Real-time look-ahead trajectory planning algorithm based on small line segment pose synchronization,” in *2025 10th Asia-Pacific Conference on Intelligent Robot Systems (ACIRS)*, 2025, pp. 312–318.
- [87] T. Faulwasser, T. Weber, P. Zometa, and R. Findeisen, “Implementation of nonlinear model predictive path-following control for an industrial robot,” *IEEE Transactions on Control Systems Technology*, vol. 25, no. 4, pp. 1505–1511, 2017, arXiv:1506.09084.
- [88] H. Sun and H. Song, “Online predictive motion control of redundant robots via model predictive control and full null space parameterization,” in *2024 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2024, pp. 1501–1506.
- [89] X. Wang, H. Li, D. Wang, X. Huang, and Z. Jiang, “Constrained sampling-based MPC using path integral for collision-free robot manipulation,” TechRxiv preprint (not peer reviewed), 2024.
- [90] J. Wang, S. Sharifi, and M. Fazlyab, “Generalized model predictive path integral control as expectation–maximization,” *IEEE Control Systems Letters*, vol. 10, pp. 2119–2124, 2026, arXiv:2606.00317.
- [91] N. Gautier, Y. Guillemit, Y. Sebsadji, M. Porez, and D. Chablat, “Comparison of robot morphologies and base positioning for welding applications,” in *Proc. ASME Int. Design Engineering Technical Conf. (IDETC)*, 48th Mechanisms and Robotics Conference, vol. 7, 2024, arXiv:2407.05720.
- [92] Q. Tang, L. Ma, D. Zhao, Y. Sun, and Q. Wang, “A dual-robot cooperative arc welding path planning algorithm based on multi-objective optimization,” *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 3048–3053, 2023.
- [93] A. Razjigaev, H. Lohr, A. Vargas-Uscategui, P. King, and T. Bandyopadhyay, “Fast functionally redundant inverse kinematics for robotic toolpath optimisation in manufacturing tasks,” arXiv preprint arXiv:2512.10116, 2025. [Online]. Available: <https://arxiv.org/abs/2512.10116>

- [94] X. Ye, M. Grobbel, T. Schürmann, S. Schwab, and S. Hohmann, "Path-constrained trajectory planning for multi-robot manufacturing systems using null-space descent optimization with reduced hessian," *The International Journal of Robotics Research*, vol. 45, no. 11, pp. 1648–1681, 2026.
- [95] J. Yang, Q. Qi, A. Adili, and H. Ding, "An analytical tool path smoothing algorithm for robotic machining with the consideration of redundant kinematics," *Robotics and Computer-Integrated Manufacturing*, vol. 89, p. 102768, 2024.
- [96] L. Lu, L. Zhang, C. Fan, and H. Wang, "High-order joint-smooth trajectory planning method considering tool-orientation constraints and singularity avoidance for robot surface machining," *Journal of Manufacturing Processes*, vol. 80, pp. 789–804, 2022.
- [97] Z. Yin, F. Wu, Z. Wang, J. Yang, J. Tan, and D. Kong, "Dynamic programming-based offline redundancy resolution of redundant manipulators along prescribed paths with real-time adjustment," arXiv preprint arXiv:2411.17052, 2024. [Online]. Available: <https://arxiv.org/abs/2411.17052>
- [98] A. G. Dharmawan, S. Padmanathan, Y. Xiong, I. F. Ituarte, S. Foong, and G. S. Soh, "Maximizing robot manipulator's functional redundancy via sequential informed optimization," in *2018 3rd International Conference on Advanced Robotics and Mechatronics (ICARM)*, 2018, pp. 334–339.
- [99] M. FarzanehKaloorazi, I. A. Bonev, and L. Birglen, "Simultaneous path placement and trajectory planning optimization for a redundant coordinated robotic workcell," *Mechanism and Machine Theory*, vol. 130, pp. 346–362, 2018.
- [100] J. De Maeyer and E. Demeester, "Benchmarking framework for robotic arc welding motion planning," *Procedia CIRP*, vol. 97, pp. 247–252, 2021.
- [101] M. Fazlyab, S. Sharifi, and J. Wang, "Model predictive path integral control as preconditioned gradient descent," *IEEE Control Systems Letters*, vol. 10, pp. 2089–2094, 2026.